



Figure 8: Iceweasel works a bit slow on Yeeeloong 8089B

people allocated to the project is far from desirable. Neither popular browser (Chromium or Firefox) has been ported to MIPS. We can hope FOSS developers will do so, eventually. Practically, it depends on the availability of desktop MIPS systems. Developers are humans; when you donate your hardware to them, you can raise interest in porting. Anyway, almost all general-purpose code is already being compiled for the MIPSEL architecture, so feel free to upgrade your Debian.

Among software already compiled for MIPS is the Firefox fork 'Iceweasel'. At version 3.5.16, it's a bit behind the current Firefox 9.0. There is a Qt-based browser called Arora, which can be installed off the repo, or compiled from scratch. However, having a slow processor makes it difficult to comfortably surf the Web.

Well, so what are the new features that have appeared in Debian Squeeze? In short: improved power management—the CPU can scale down to 200MHz when there's no load. Flash videos now can be decoded by Gnash. At least it was stated so; I've actually never seen it. Kernel support was tuned a bit, too.

To get Flash player, we need to take a few extra steps. Because neither Adobe, nor MIPS Inc., nor even Debian have downloads for Flash-player. That's why I have to find `libflashplayer.so` file. Again, I was lucky. And again, it resides inside Android tablet—but this time, inside Ainol Novo7 Basic. I've exported it and I tried to start Firefox with that plug-in library.

```
loongson@debian:~$ firefox
LoadPlugin: failed to initialize shared library /usr/lib/mozilla/
plugins/libflashplayer.so [/usr/lib/libc.so: ELF file version does
not match current one]
ERROR: Too many attempts to read from the player!
```

Alas! The trick that worked for the TrimSlice was unsuccessful this time. Apparently, the netbook and Ainol use two different MIPS processors, of different series.

```
loongson@debian:~$ file /usr/lib/mozilla/plugins/libflashplayer.so
/usr/lib/mozilla/plugins/libflashplayer.so: ELF 32-bit LSB shared
object, MIPS, MIPS32 version 1 (SYSV), dynamically linked, with
unknown capability 0xf41 = 0x756e6700, with unknown capability
0x78100 = 0x18400000, stripped
```

```
loongson@debian:~$ file /usr/lib/gnash/libgnashplugin.so
/usr/lib/gnash/libgnashplugin.so: ELF 32-bit LSB shared object,
MIPS, MIPS-I version 1 (SYSV), dynamically linked, with unknown
capability 0xf41 = 0x756e6700, with unknown capability 0x78100 =
0x18400000, stripped
```

Our hypothesis is partly correct. The 'native' Gnash application (uses MIPS-I standard) is correct, but it doesn't work. The 'imported' `libflashplayer.so` is believed to be compatible with MIPS, but is for another MIPS (MIPS32) family, and has the wrong dependencies for the `libc` library. I read the tablet's specifications and figured out that the Ainol tablet is built on an Ingenix JZ4770 Xburst CPU, which has some differences from the Loongson2F.

Iceweasel passed the ACID3 and SunSpider tests, scoring 98 out of 100 for ACID3, but showing an extremely low result for SunSpider—almost 10 times slower than Chromium/Firefox on ARM. As we can see, CPU performance is a bottleneck again.

In general, all software packages in the Debian/MIPSEL port are present. The main task for the manufacturer is to increase the speed of the MIPS processor to convert it into a faster desktop machine, so QEMU would report (for a virtual Windows XP installation—yes, QEMU on MIPS works smoothly) a quicker CPU, not the currently available 41 MHz (!) by Intel.

The fact that there are now more and more products based on alternative (non-x86) processors is very promising. Even more interesting is the fact that they are designed to be efficient in terms of energy consumption. Both ARM and MIPS showed the same consumption—a couple of watts. Moreover, the trend to integrate a hardware decoding/encoding chip into the CPU became so popular, we're currently facing an outstanding moment—systems with quite ordinary central processors, yet with a brilliant capability to hardware-accelerate video- and audio-streams (like the Tegra2 processor). The chances for ARM/MIPS products to capture the conservative desktop market are enormous. Intel, your turn! 

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By: Anton Borisov

The author has specialised in Linux and FOSS technologies for more than a decade. His professional spheres of interest include, but are not limited to, robotics, embedded systems, statistics and algorithmic methods.